1. Down's Syndrome/Trisomy – 21

Trisomic condition of chromosome - 21 results in Down's syndrome. It is characterized by severe mental retardation, defective development of the central nervous system, increased separation between the eyes, flattened nose, ears are malformed, mouth is constantly open and the tongue protrudes.

2. Patau's Syndrome/Trisomy-13

Trisomic condition of chromosome 13 results in Patau's syndrome. Meiotic non disjunction is thought to be the cause for this chromosomal abnormality. It is characterized by multiple and severe body malformations as well as profound mental deficiency. Small head with small eyes, cleft palate, malformation of the brain and internal organs are some of the symptoms of this syndrome.

b. Allosomal abnormalities in human beings

Mitotic or meiotic non-disjunction of sex chromosomes causes allosomal abnormalities. Several sex chromosomal abnormalities have been detected. Eg. Klinefelter's syndrome and Turner's syndrome.

1. Klinefelter's Syndrome (XXY Males)

This genetic disorder is due to the presence of an additional copy of the X chromosome resulting in a karyotype of 47,XXY. Persons with this syndrome have 47 chromosomes (44AA+XXY). They are usually sterile males, tall, obese, with long limbs, high pitched voice, under developed genitalia and have feeble breast (gynaecomastia) development.

2. Turner's Syndrome (XO Females)

This genetic disorder is due to the loss of a X chromosome resulting in a karyotype of 45,X. Persons with this syndrome have 45 chromosomes (44 autosomes and one X chromosome) (44AA+XO) and are sterile females. Low stature, webbed neck, under developed breast, rudimentary gonads lack of menstrual cycle during puberty, are the main symptoms of this syndrome.

Summary

Genetics is a branch of biology that deals with the study of heredity and variation. It describes how characteristics and features pass on from the parents to their offsprings in successive generations. Variation is the degree by which progeny differ from their parents. A set of three or more alleles of the same gene occupying the same locus in a given pair of homologous chromosomes controlling a particular trait is called Multiple allele. ABO blood grouping in man is a good example for multiple allelism. Apart from A and B antigens, the RBC's of humans contain a special type of antigen called Rh antigen/Rh factors. Erythroblastosis foetalis, also called haemolytic disease of the newborn, in which the red blood cells of a foetus are destroyed due to maternal immune reaction resulting from a blood group incompatibility between the foetus and the mother.

The mechanism of determination of male and female individuals in a species is called sex determination. The chromosomes are different in two sexes and referred to as allosomes; the remaining chromosomes are named autosomes. The inheritance of a trait that is determined by a gene located on one of the sex chromosomes is called sex linked inheritance. Haemophilia, colourblindness, muscular dystrophy are some examples for X linked inheritance in human beings.

Pedigree analysis is the study of traits as they have appeared in a given family line for several generations. The genetic disorders are of two types- Mendelian and chromosomal. Alternations or mutation in single gene causes Mendelian disorders like, thalassemia, albinism, phenylketonuria, and Huntington's chorea. Chromosomal abnormalities arise due to chromosomal non-disjunction, translocation, deletion, duplication and inversion. Downs syndrome, Klinefelter's syndrome, Turner's syndrome and Patau's syndrome are some of the chromosomal disorders. Downs syndrome is due to trisomy of chromosome 21. Presence of trisomic condition of chromosome 13 results in

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